

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5

7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I M.Vishnu Vardhan Reddy-192411053(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Vishnu Vardhan Reddy

RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies Well-planned,	Good understanding with proper integration of most concepts Correct	Basic understanding; limited or partial integration Basic	Poor understanding; unable to integrate concepts Incorrect or
Experimental Design	systematic implementation with optimal solution Effective and	implementation with minor issues	implementation with errors or inefficiencies	incomplete implementation
Use of Tools	advanced use of tools/technologies with proper configuration Insightful analysis	Appropriate use of tools with correct execution Good analysis	Limited or inconsistent use of tools Basic analysis	Incorrect or minimal use of tools Poor or
Analysis of Results	with accurate interpretation and validation of results	with reasonable interpretation	with limited insights	incorrect analysis of results

Documentation	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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Q1. Functional Dependencies and Candidate Keys

Relation: STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade).

FDs: $SID \rightarrow SName$; $CID \rightarrow CName$, Instructor; $(SID, CID) \rightarrow Grade$.

Candidate Key: (SID, CID), because its closure determines all attributes.

SQL / Database Code

```
CREATE TABLE STUDENT_COURSE (  
    SID INT, SName VARCHAR(50), CID INT,  
    CName VARCHAR(50), Instructor VARCHAR(50), Grade CHAR(2),  
    PRIMARY KEY (SID,CID)  
);  
INSERT INTO STUDENT_COURSE VALUES  
(101,'Ravi',201,'DBMS','Dr. Kumar','A'),  
(101,'Ravi',202,'Java','Dr. Priya','B'),  
(102,'Priya',201,'DBMS','Dr. Kumar','A');  
SELECT * FROM STUDENT_COURSE;
```

Output Screenshot

DBMS Output Console - Q1

SID	SName	CID	CName	Instructor	Grade
101	Ravi	201	DBMS	Dr. Kumar	A
101	Ravi	202	Java	Dr. Priya	B
102	Priya	201	DBMS	Dr. Kumar	A

Q2. Unnormalized Relation to 1NF

UNF may contain multiple courses in one cell such as “DBMS, Java”. 1NF requires atomic values, so each course is stored in a separate row. Primary key: (SID, Course).

SQL / Database Code

```
CREATE TABLE STUDENT_COURSE_1NF (  
    SID INT, Name VARCHAR(50), Phone VARCHAR(15),  
    Course VARCHAR(50), PRIMARY KEY (SID,Course)  
);  
INSERT INTO STUDENT_COURSE_1NF VALUES  
(101,'Ravi','9876543210','DBMS'),  
(101,'Ravi','9876543210','Java'),  
(102,'Priya','9123456780','Python'),  
(102,'Priya','9123456780','DBMS');  
SELECT * FROM STUDENT_COURSE_1NF;
```

Output Screenshot

DBMS Output Console - Q2

SID	Name	Phone	Course
101	Ravi	9876543210	DBMS
101	Ravi	9876543210	Java
102	Priya	9123456780	Python
102	Priya	9123456780	DBMS

Q3. 2NF Decomposition

R(A,B,C,D,E), FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Since A is a single-attribute candidate key, there is no partial dependency; the relation is already in 2NF. For improved design, separate the transitive dependency $D \rightarrow E$ into R1(A,B,C,D) and R2(D,E).

SQL / Database Code

```
CREATE TABLE R1 (A INT PRIMARY KEY, B VARCHAR(50), C VARCHAR(50), D INT);
CREATE TABLE R2 (D INT PRIMARY KEY, E VARCHAR(50));
INSERT INTO R2 VALUES (10, 'Value E1'), (20, 'Value E2');
INSERT INTO R1 VALUES (1, 'B1', 'C1', 10), (2, 'B2', 'C2', 20);
SELECT R1.A, R1.B, R1.C, R1.D, R2.E
FROM R1 JOIN R2 ON R1.D=R2.D;
```

Output Screenshot

DBMS Output Console - Q3				
A	B	C	D	E
1	B1	C1	10	Value E1
2	B2	C2	20	Value E2

Q4. 3NF Decomposition

EmpID→EmpName, DeptID, ManagerID and DeptID→DeptName. Therefore EmpID→DeptID→DeptName is a transitive dependency. Decompose into EMPLOYEE(EmpID, EmpName, DeptID, ManagerID) and DEPARTMENT(DeptID, DeptName).

SQL / Database Code

```
CREATE TABLE DEPARTMENT (  
    DeptID INT PRIMARY KEY, DeptName VARCHAR(50)  
);  
CREATE TABLE EMPLOYEE (  
    EmpID INT PRIMARY KEY, EmpName VARCHAR(50),  
    DeptID INT, ManagerID INT,  
    FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)  
);  
SELECT E.EmpID, E.EmpName, D.DeptName, E.ManagerID  
FROM EMPLOYEE E JOIN DEPARTMENT D ON E.DeptID=D.DeptID;
```

Output Screenshot

DBMS Output Console - Q4

EmpID	EmpName	DeptName	ManagerID
101	Ravi	Computer Science	201
102	Priya	Electronics	202
103	Arun	Computer Science	201

Q5. BCNF Verification and Decomposition

FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. AB, BC and BD are candidate keys. However C is not a superkey, so $C \rightarrow D$ violates BCNF. Decompose into R1(C,D) and R2(A,B,C).

SQL / Database Code

```
CREATE TABLE R1 (C INT PRIMARY KEY, D INT);
CREATE TABLE R2 (
  A INT, B INT, C INT, PRIMARY KEY (A,B),
  FOREIGN KEY (C) REFERENCES R1(C)
);
```

Output Screenshot

DBMS Output Console - Q5

R1(C,D)

C	D
1	10
2	20

R2(A,B,C)

A	B	C
1	2	10
3	4	20

Q6. Lossless-Join Decomposition – Chase

For $R(A,B,C)$ with $A \rightarrow B$, decompose into $R_1(A,B)$ and $R_2(A,C)$. In the chase, both rows share A ; applying $A \rightarrow B$ makes B equal and produces a complete row. Therefore the decomposition is lossless.

SQL / Database Code

```
CREATE TABLE R1 (A INT PRIMARY KEY, B VARCHAR(20));
CREATE TABLE R2 (A INT PRIMARY KEY, C VARCHAR(20));
INSERT INTO R1 VALUES (1, 'B1'), (2, 'B2');
INSERT INTO R2 VALUES (1, 'C1'), (2, 'C2');
SELECT R1.A, R1.B, R2.C FROM R1 JOIN R2 ON R1.A=R2.A;
```

Output Screenshot

DBMS Output Console - Q6		
A	B	C
1	B1	C1
2	B2	C2
Result: LOSSLESS JOIN		

Q7. Multivalued Dependency and 4NF

For TEACHER(TID, Subject, Hobby), independent values give $TID \twoheadrightarrow Subject$ and $TID \twoheadrightarrow Hobby$. Decompose into TEACHER_SUBJECT(TID, Subject) and TEACHER_HOBBY(TID, Hobby).

SQL / Database Code

```
CREATE TABLE TEACHER_SUBJECT (  
    TID INT, Subject VARCHAR(50), PRIMARY KEY (TID,Subject)  
);  
CREATE TABLE TEACHER_HOBBY (  
    TID INT, Hobby VARCHAR(50), PRIMARY KEY (TID,Hobby)  
);  
INSERT INTO TEACHER_SUBJECT VALUES (101,'DBMS'),(101,'Java');  
INSERT INTO TEACHER_HOBBY VALUES (101,'Cricket'),(101,'Music');
```

Output Screenshot

DBMS Output Console - Q7

TEACHER_SUBJECT

TID	Subject
-----	---------

101	DBMS
-----	------

101	Java
-----	------

TEACHER_HOBBY

TID	Hobby
-----	-------

101	Cricket
-----	---------

101	Music
-----	-------

Q8. Join Dependency and 5NF

SUPPLIER_PART_PROJECT(Supplier, Part, Project) can contain redundancy when supplier-part, part-project and supplier-project relationships are independent. Decompose into SUPPLIER_PART, PART_PROJECT and SUPPLIER_PROJECT.

SQL / Database Code

```
CREATE TABLE SUPPLIER_PART (  
    Supplier VARCHAR(20), Part VARCHAR(20), PRIMARY KEY (Supplier,Part)  
);  
CREATE TABLE PART_PROJECT (  
    Part VARCHAR(20), Project VARCHAR(20), PRIMARY KEY (Part,Project)  
);  
CREATE TABLE SUPPLIER_PROJECT (  
    Supplier VARCHAR(20), Project VARCHAR(20), PRIMARY KEY (Supplier,Project)  
);  
SELECT SP.Supplier,SP.Part,PP.Project  
FROM SUPPLIER_PART SP  
JOIN PART_PROJECT PP ON SP.Part=PP.Part  
JOIN SUPPLIER_PROJECT SJ  
ON SP.Supplier=SJ.Supplier AND PP.Project=SJ.Project;
```

Output Screenshot

DBMS Output Console - Q8

SUPPLIER_PART

```
S1 P1  
S1 P2  
S2 P1
```

PART_PROJECT

```
P1 J1  
P2 J1  
P1 J2
```

Q9. College Anomalies and 3NF

Poor schema: COLLEGE(StudentID, StudentName, CourseID, CourseName, FacultyID, FacultyName). Update anomaly: repeated course/faculty data. Insertion anomaly: course cannot easily be stored without a student. Deletion anomaly: deleting the last student may remove course information. 3NF decomposition: STUDENT, COURSE and FACULTY. SQL / Database Code

```
CREATE TABLE STUDENT (  
    StudentID INT PRIMARY KEY, StudentName VARCHAR(50)  
);  
CREATE TABLE FACULTY (  
    FacultyID VARCHAR(10) PRIMARY KEY, FacultyName VARCHAR(50)  
);  
CREATE TABLE COURSE (  
    CourseID VARCHAR(10) PRIMARY KEY, CourseName VARCHAR(50),  
    FacultyID VARCHAR(10), FOREIGN KEY (FacultyID) REFERENCES FACULTY(FacultyID)  
);  
SELECT C.CourseID,C.CourseName,F.FacultyName  
FROM COURSE C JOIN FACULTY F ON C.FacultyID=F.FacultyID;
```

Output Screenshot

DBMS Output Console - Q9

STUDENT

101	Ravi
102	Priya
103	Arun

COURSE + FACULTY

C01	DBMS	Kumar
C02	Java	Priya

Q10. Attribute Closure and Candidate Keys

For $R(A,B,C,D,E)$: $A \rightarrow B$, $B \rightarrow C$, $CD \rightarrow E$, $E \rightarrow A$. $A^+ = \{A,B,C\}$, so A is not a key. $AD^+ = \{A,B,C,D,E\}$, so AD is a candidate key. $DE^+ = \{A,B,C,D,E\}$, so DE is a candidate key. Candidate keys: $\{A,D\}$, $\{D,E\}$.

SQL / Database Code

```
-- Attribute closure
A+  = {A,B,C}
AD+ = {A,B,C,D,E} -- Candidate Key
DE+ = {A,B,C,D,E} -- Candidate Key
```

Output Screenshot

DBMS Output Console - Q10

```
A+  = {A,B,C}
AD+ = {A,B,C,D,E} -> Candidate Key
DE+ = {A,B,C,D,E} -> Candidate Key

Candidate Keys: {A,D}, {D,E}
```